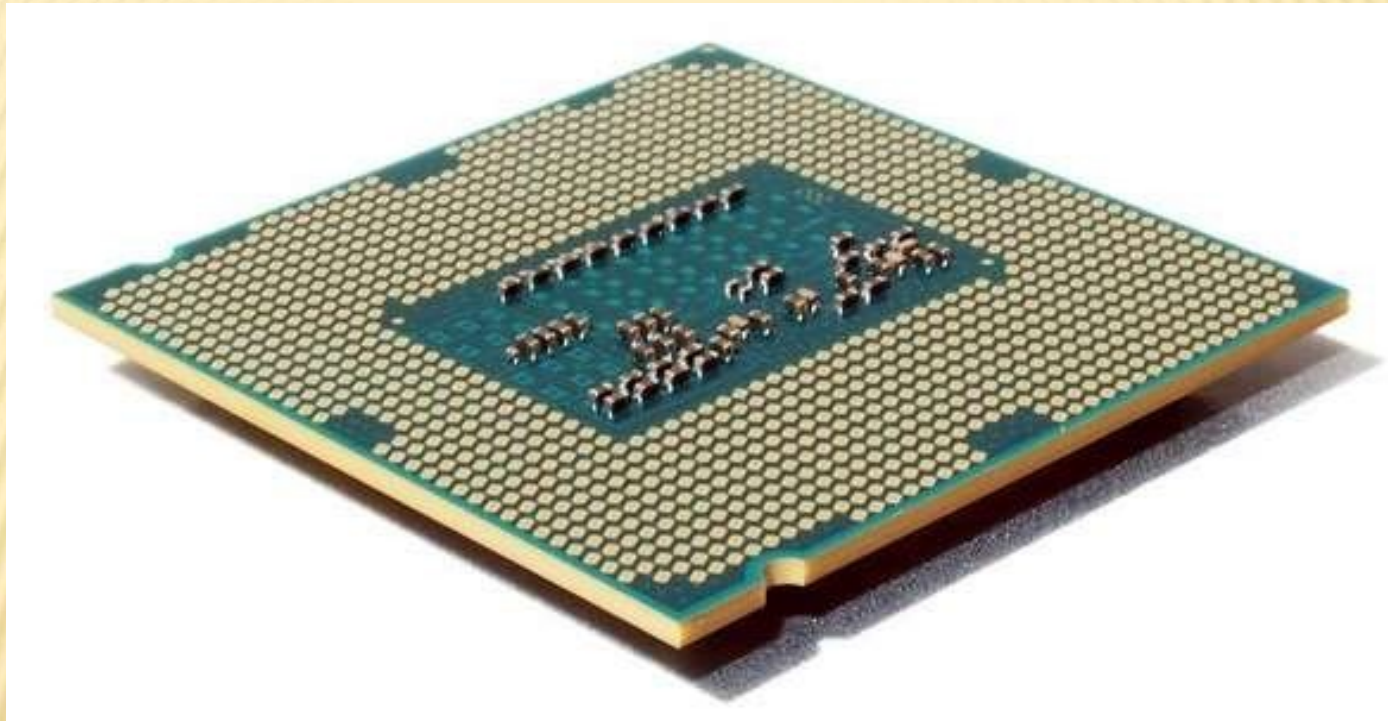


Processors



Outline:

1. What is a Processor?
2. Moore's Law
3. History
4. Types of Processors
5. Performance
6. Need Of Processor
7. Functions
8. Processor memory types
9. Advantages
10. Disadvantages
11. Conclusion

What is a Processor?

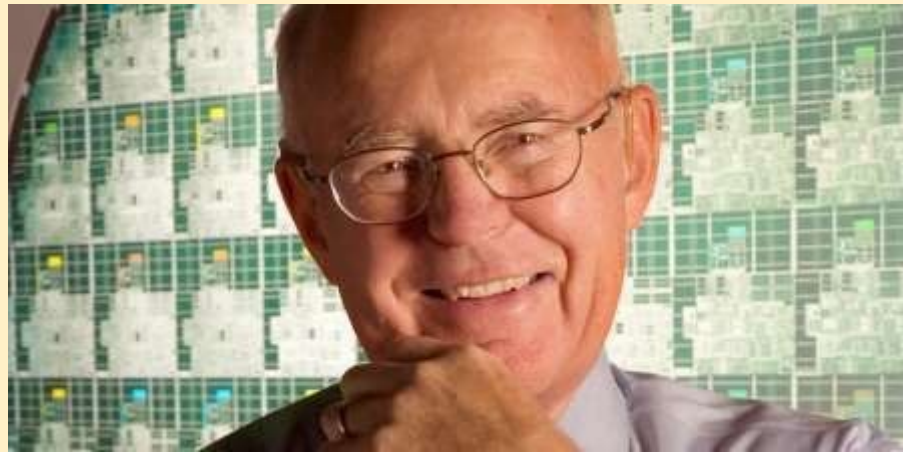
A processor is a small chip that resides in the computers. Its basic job is to receive input and the provide the appropriate output.



The term processor has generally replaced the term Central processing unit (CPU).

The processor in a personal computer or in small devices is often called a Microprocessor.

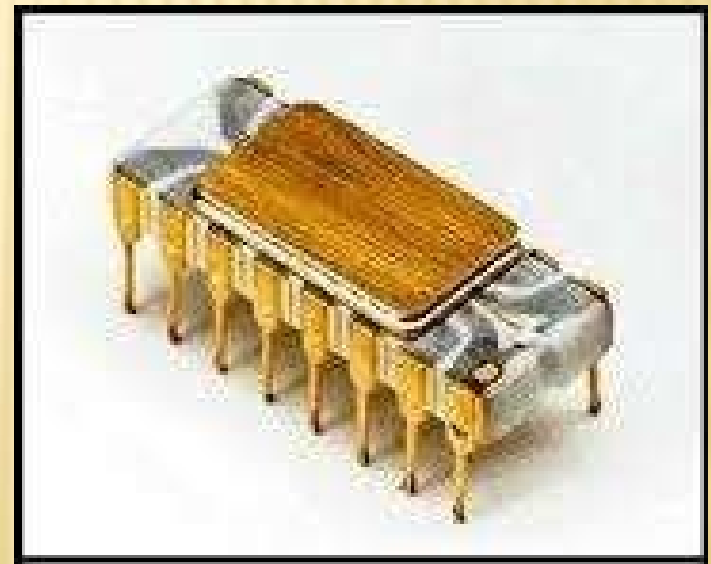
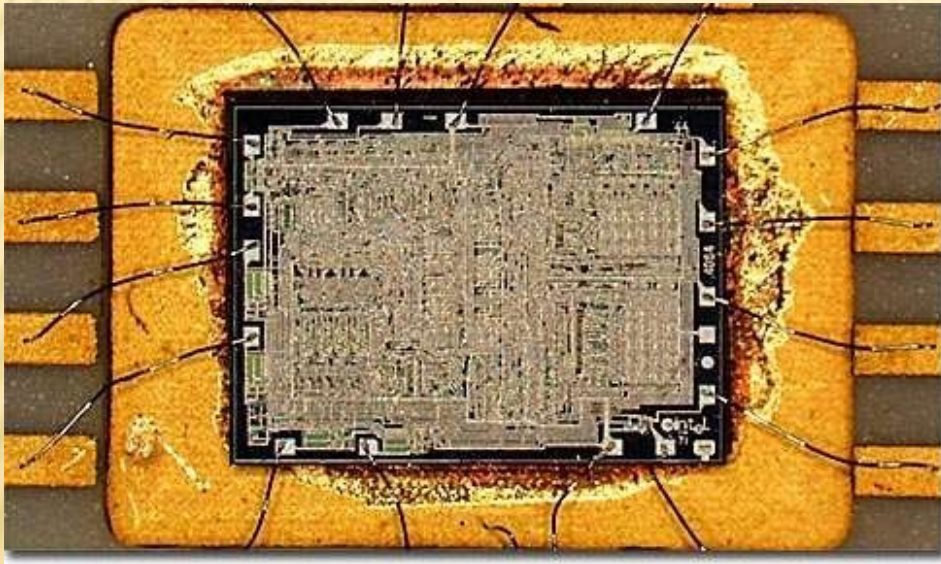
MOORE'S LAW



- Moore's law is the observation that, over the history of computing hardware, the number of transistors in a dense integrated circuit doubles approximately every two years. The law is named after Gordon E. Moore, co-founder of the Intel Corporation, who described the trend in his 1965 paper. His prediction has proven to be accurate, in part because the law is now used in the semiconductor industry to guide long-term planning and to set targets for research

History:

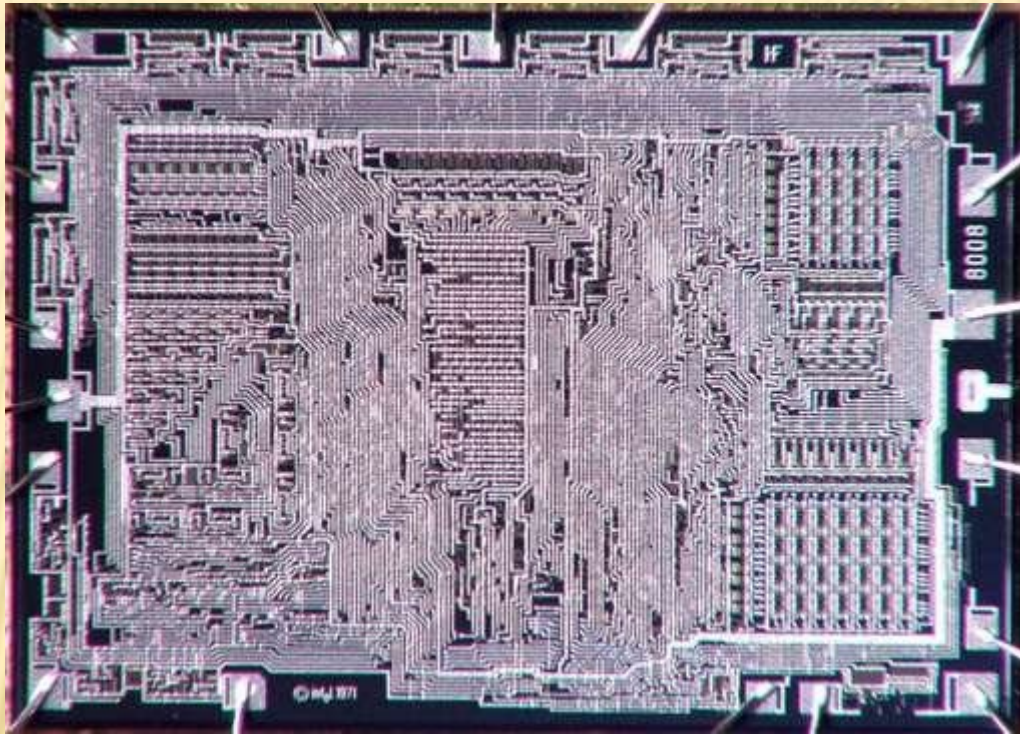
The first microprocessors emerged in the early 1970s and were used for electronic calculators, using binary-coded decimal (BCD) arithmetic on 4-bit words.



Intel introduced its first 4-bit microprocessor 4004 in 1971

History:

The Intel 8008 was introduced in 1972 ,the world's first 8-bit microprocessor.



But it had some performance issues, So Intel invented another 8-bit microprocessor named as Intel 8080 in 1974

Types:

1. Single Core Processor:

It is a processor that has only one core, so it can only start one operation at a time. It can however in some situations start a new operation before the previous one is completed. Originally all processors were single core. Multi Core processors were introduced later.



- Apple single-core A5 chip

Types:

2. Multi-core or Many core Processor:

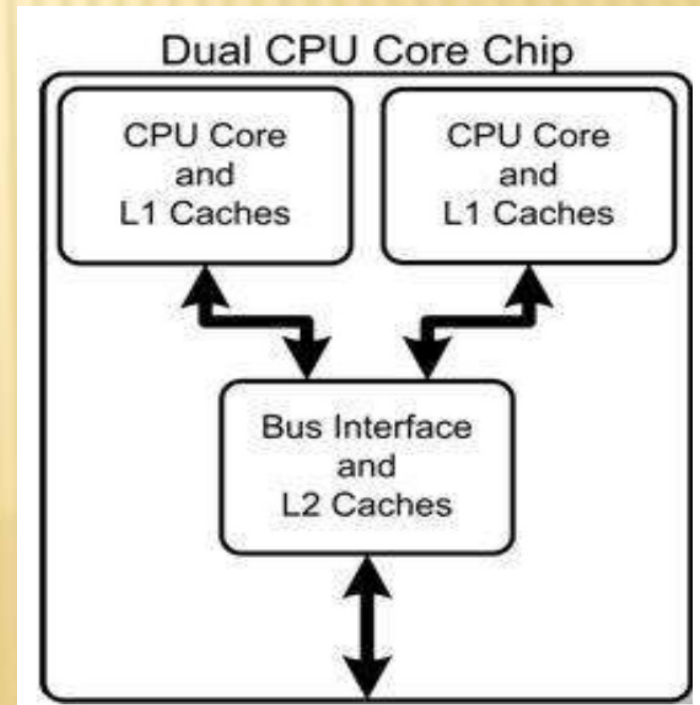
A **multi-core processor** is a single computing component with two or more independent actual central processing units (called "cores"), which are the units that read and execute program instructions.



Types of Multi-Core:

1. Dual Core Processor:

It has combined two processors onto a single **integrated circuit**. These processors are well-suited for [multitasking](#) because there are two complete execution [cores](#) instead of one.



2. Quad-Core Processor:

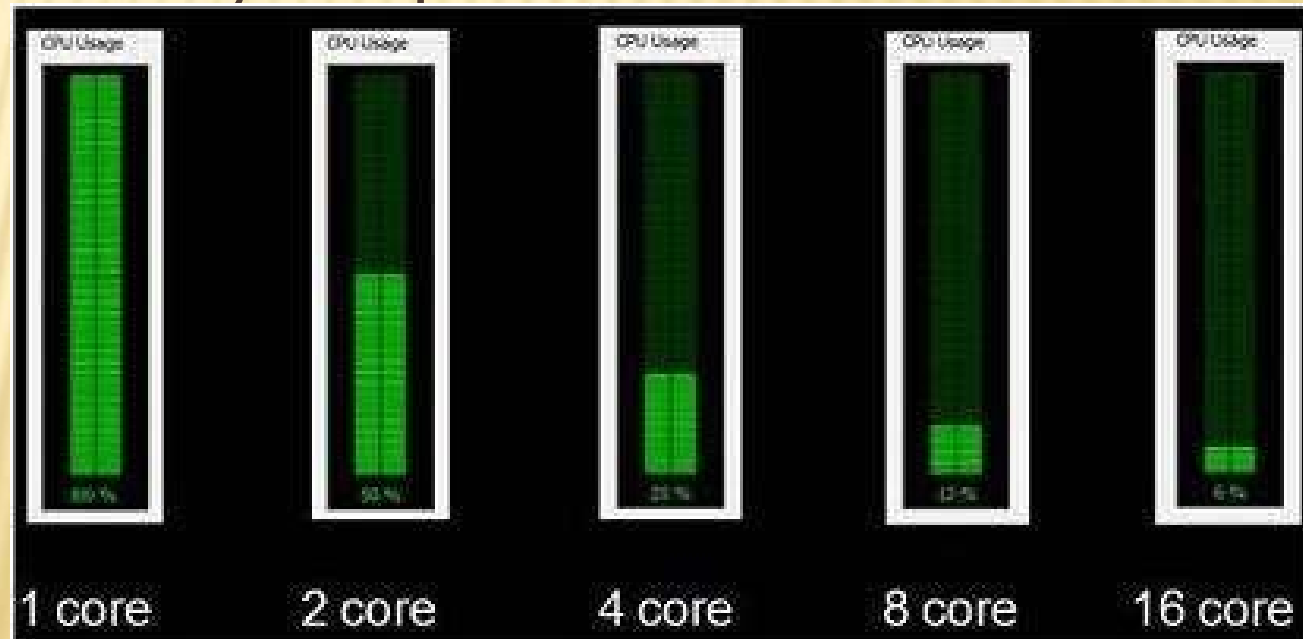
A quad-core processor is a chip with four independent units called cores that read and execute central processing unit (CPU) instructions such as add, move data, and branch.



Quad-core and higher multi-core processor configurations have become common for general-purpose computing, not only for PCs but for mobile devices such as smartphones and tablets.

PERFORMANCE

- ✗ The clock speed-also know as clock rate- indicates how fast the CPU can run. This is measured in megahertz (MHz) or gigahertz (GHz) and corresponds with how many instruction cycles the CPU can deal with in a second. A 2 GHz CPU performs two billion cycles per second.



Performance

- PERFORMANCE OF COMPUTERS IS INCREASED BY USING MULTI-CORE PROCESSORS, WHICH ESSENTIALLY IS PLUGGING TWO OR MORE INDIVIDUAL PROCESSORS (CALLED CORES IN THIS SENSE) INTO ONE INTEGRATED CIRCUIT. IDEALLY, A DUAL CORE PROCESSOR WOULD BE NEARLY TWICE AS POWERFUL AS A SINGLE CORE PROCESSOR. INCREASING THE NUMBER OF CORES IN A PROCESSOR INCREASES THE WORKLOAD THAT CAN BE HANDLED.

Need Of Multi-core Processors:

The basic need of multi-core processors is the *Parallel Computing*. Time consumption is the pulling drawback of Single core processors. So, Multi-core technology has become the mean to achieve efficiency through parallel processing.



Functions

- Also known as Information Processing Cycle/
Processing Cycle/Machine Processing Cycle.
- has four primary functions:
 1. **fetch** (program instructions from RAM as input)
 2. **decode** (convert the instructions to its understandable form)
 3. **execute** (processes it)
 4. **write back** (sends back the computed results)

Processor Memory Types:

- **Shared memory:**

In this model, there is one (large) common shared memory for all processors.

- **Distributed memory:**

In this model, each processor has its own (small) local memory, and its content is not replicated anywhere else

Why Multi-core?

- **Better Performance**

- For the *Multi tasking*

- e.g. Burning CD with graphic works at the same time

- **Economical efficiency**

- A dual-core is much cheaper than two single cores

Why Multi-core?

- **Save the room of motherboard**

- Two single cores → In one die
 - We can use this room more efficiently

- **Parallel Computing**

- As every core executes instructions as individual processing unit, parallel processing is Achieved

ADVANTAGES

❖ **Speed**

- ❑ It is measured in hertz.
- ❑ For instance, a microprocessor with 3 GHz (means 3 billion tasks per second).

❖ **Data Movement**

- ❑ Quickly movement of data between the various memory locations.

❖ **Multi-tasking productivity.**

DISADVANTAGES

- ❖ Some of the disadvantages with Microprocessor are that it might get overheated, and the limitation it imposes on the size of data.
- ❖ If using dual core, they do not work at twice the speed as a normal processor. They get only **60-80%** more speed.

Conclusion:

- All computers are now parallel computers.
- Multi-core processors represent an important new trend in computer architecture.
 - Decreased power consumption and Better Performance.
 - Efficient Processing.
- They enable true thread-level parallelism with great energy efficiency.
- To utilize their full potential, applications will need to move from a single to a multi-threaded model.

Thank You.!!